

NAKHON RATCHASIMA, Thailand

WMO: 484310

Lat: 14.968N

Lon: 102.086E

Elev: 614

StdP: 14.37

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	60.5	63.2	49.6	53.8	68.5	52.3	59.5	70.2	12.1	76.1	11.0	76.3	1.8	60	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.1	100.6	77.6	98.6	77.7	96.8	77.6	81.5	92.2	80.8	91.6	80.2	90.6	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	154.4	85.1	78.3	150.4	84.4	77.8	147.9	84.0	45.7	92.4	45.1	91.9	44.4	91.1	93.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.1	9.2	7.8	WB	55.9	103.3	2.8	2.1	53.8	104.8	52.2	106.1	50.6	107.3	48.5	108.8	
				51.8	84.5	2.7	2.9	49.9	86.6	48.3	88.3	46.8	89.9	44.9	92.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.7	77.0	81.1	85.0	87.3	86.3	86.0	84.9	84.2	82.8	81.7	79.4	76.4
	DBStd	4.94	4.70	5.02	4.39	3.50	2.97	2.27	2.30	2.09	2.12	2.34	3.57	4.74
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	0	0	0	0	0	0	0	0	0	0	0	1
	CDD50	11924	836	872	1086	1118	1126	1079	1082	1061	983	981	883	817
	CDD65	6450	371	452	621	668	661	629	617	596	533	516	433	353
	CDH74	73224	3488	5126	7836	8679	8121	7816	7324	6751	5519	5254	4158	3152
CDH80	33179	1384	2504	4193	4744	3984	3745	3286	2824	2030	1866	1512	1107	
Wind	WSAvg	2.4	2.2	2.1	2.1	2.0	2.2	2.8	2.9	2.7	1.9	2.4	2.9	2.9
Precipitation	PrecAvg	41.8	0.2	0.6	1.8	2.9	5.8	4.2	5.0	5.9	9.5	5.7	1.1	0.1
	PrecMax	54.6	2.4	3.7	9.8	10.0	12.6	11.3	11.5	12.8	21.5	12.7	5.2	1.3
	PrecMin	23.6	0.0	0.0	0.0	0.3	1.3	0.0	0.4	1.9	2.4	0.0	0.0	0.0
	PrecStd	8.2	0.4	0.8	1.7	2.1	2.6	2.4	2.3	2.7	3.5	3.2	1.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.5	99.0	102.5	103.9	101.3	98.7	97.6	95.5	93.6	92.8	94.3	93.3
		MCWB	73.6	75.6	75.7	77.5	79.8	79.2	79.1	78.6	79.8	78.5	76.9	75.8
	2%	DB	92.3	96.8	99.9	101.3	98.3	96.3	95.0	93.7	91.9	90.8	91.5	90.8
		MCWB	73.8	75.2	75.8	78.0	79.5	78.9	78.6	78.5	79.3	77.7	76.6	74.8
	5%	DB	90.1	94.7	97.8	98.8	95.9	94.5	93.2	92.2	90.4	89.4	89.5	88.4
		MCWB	73.5	74.9	76.1	78.4	79.3	78.7	78.2	78.3	78.8	77.4	75.7	73.6
	10%	DB	87.5	92.2	95.4	96.4	93.7	92.7	91.5	90.5	88.9	88.0	87.4	86.1
		MCWB	72.8	74.7	76.2	78.3	79.0	78.6	78.0	78.1	78.3	76.9	74.8	72.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.9	80.1	80.7	81.8	82.2	81.7	81.4	81.6	82.2	80.7	79.5	77.9
		MCDB	89.1	93.1	93.8	95.8	94.7	92.4	91.6	90.2	90.0	88.2	88.9	89.5
	2%	WB	76.6	78.2	79.2	80.7	81.1	80.7	80.2	80.5	80.8	79.4	78.3	76.5
		MCDB	86.8	90.7	93.0	94.9	92.9	91.4	90.1	89.4	88.7	86.7	87.1	87.1
	5%	WB	75.4	77.1	78.3	79.9	80.5	80.0	79.5	79.7	79.9	78.8	77.5	75.3
		MCDB	85.4	89.4	91.9	93.4	91.8	90.3	89.2	88.2	87.5	86.1	85.7	85.4
	10%	WB	74.2	76.0	77.5	79.1	79.8	79.3	78.8	78.9	79.1	78.1	76.5	74.0
		MCDB	84.0	87.8	90.4	91.9	90.2	89.1	88.4	87.2	86.2	84.9	84.1	83.6
Mean Daily Temperature Range	5% DB	MDBR	20.0	20.7	19.6	18.1	15.1	14.3	13.8	13.2	12.4	12.4	14.8	17.7
		MCDBR	21.9	22.9	22.9	20.9	17.7	16.3	15.9	15.2	14.4	14.0	16.9	19.1
	5% WB	MCWBR	6.7	5.8	5.1	4.7	4.6	4.0	4.2	4.1	4.4	3.9	4.9	6.0
		MCDWR	18.0	19.8	19.6	18.6	15.8	14.7	14.5	13.8	13.5	12.8	14.8	17.6
Clear-Sky Solar Irradiance	taub		0.494	0.600	0.646	0.592	0.499	0.462	0.451	0.455	0.491	0.533	0.482	0.469
	taud		2.042	1.791	1.725	1.892	2.196	2.306	2.335	2.312	2.198	2.045	2.163	2.142
	Ebn at Noon		253	231	225	238	257	264	267	269	260	246	255	256
	Edh at Noon		52	70	76	65	47	42	41	42	47	54	46	46
All-Sky Solar Radiation	RadAvg		1563	1660	1702	1818	1751	1660	1572	1528	1477	1516	1546	1529
	RadStd		100	70	84	82	102	77	93	91	81	107	82	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	+0.97	+0.86	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	+0.52	N/A	N/A	N/A	+0.36	+0.32	N/A	N/A	+225	+225

Nomenclature: See separate page